HW9.

1. Proof $N(i) \cup (i) = \left(\bigcup_{C \in C(G) \text{ and } i \in C} C \right)$

$N(i)$ denote the set of neighbors of the vertex i in the graph $G=(V, E)$, $C(G)$ represent the set of all cliques in the graph. A clique c in a graph is a subset of its vertices such that every two vertices in c are adjacent.

① Show every element in LHS is also in RHS

LHS: $N(i) \cup (i)$ RHS: the union of all cliques c that contain i.

i) i itself is clearly in in RHS, since it belongs to at least the trivial clique $\{i\}$.

ii) For any j in $N(i)$ $(j \neq i)$, j is a neighbor of i. Thus, there is at least one clique contains i and j. Thus, every neighbor j of i will also be in some cliques c that contains i, hence in the RHS.

② Show that every element in RHS is also in LHS.

By definition of cliques. every clique c that contains i is a subset of the vertices that are either i itself are are adjacent to i. If it's i itself, i is clearly in LHS. If

$j \neq i$ and j is in a clique ~~of~~ contains i, j is connected by an edge, which indicate j is a neighbor of i. with $\neq i$

Therefore, j is in LHS too.